

Predictive Structure of the Perpetual-Futures Limit Order Book

Universality, Orthogonality, and the Execution Boundary

lob_predictive_structure_preprint.tex · 8pp · Jul 2026 · measurement-first sharpening of #2 · the only paper in the set that cites inline (natbib)

Thesis

A 236-feature census of the book at 100 ms cadence (2.17 M ticks/symbol, BTC/ETH/SOL), horizons 1 s–100 min. The book carries **two empirically orthogonal predictive channels**, and their monetization is sharply bounded at the point of execution.

The factorization result (the headline)

Directional channel	rank IC up to 0.47 at 1–5 s
carried by	imbalance family, 8 independent axes
volatility info	zero
Volatility channel	peak IC 0.35 at 5 s
carried by	arrival intensity, activity counts, toxicity
directional info	zero

Disjoint feature families, on all three symbols. **Sign and magnitude of short-horizon returns are independently predictable from disjoint feature sets** — which is what licenses a two-block (sign × size) forecasting architecture instead of one joint model.

Spectral localization

The entire directional IC lives in the 0.005–0.1 Hz band. OU half-life 5–7 s. Dominant cross-coherence at 68 s. **Time-domain aggregation destroys the signal** — if you bar it up, you have thrown the alpha away before you started.

Efficiency drift, measured in real time

Half-life *shortens* May→June as spreads tighten in a maturing venue: 5 s IC 0.45 → 0.29 while 1 s holds or improves. Not a stationary edge — a decaying one, and the decay is observable while it happens. Anything built on the 5 s horizon has a shelf life.

The execution boundary

$$\begin{aligned} \text{IC} &= 0.45 \xrightarrow{\text{cond. on directionally-consistent maker fill}} 0.03 \\ \text{IC} &= 0.45 \xrightarrow{\text{cond. on the presence of any fill}} \mathbf{0.52} \end{aligned}$$

The second line is the one that matters and is unique to this paper. Any-fill conditioning *raises* IC; directional-fill conditioning collapses it. That pair isolates **adverse selection** — not signal decay, not estimation noise — as the binding constraint. Regime gating on **ent_book_shape** partially reopens it (+22%).

Novelty ledger (write related-work from this)

Known, must cite, do not re-claim: imbalance/OFI predicts short-horizon returns (Cont–Kukanov–Stoikov 2014; deep-LOB literature). New here:

1. Orthogonality factorization across 3 symbols.
2. Universality at scale — 207-feature census, 6 horizons, 3 symbols, bootstrap CIs, compressing to 8 independent directional axes.
3. Spectral localization + OU half-life.
4. Efficiency drift measured in real time.
5. The execution boundary as a *mechanism* section, not the headline.

Status

[!] Out-of-time replication on an independent month is **in progress**, not done. The paper says so in an author footnote and in the abstract; do not let a reader infer otherwise. All other

estimates come from the internal empirical record consolidated 2026-07-25 (FINDINGS.md).

Implications the paper states

Two-block (sign × size) forecasting · entropy-gated conditioning · execution-timing overlays for *slower* strategies — i.e. use the fast signal to time entries for something that isn't fast, since the fast thing itself cannot be filled profitably.

Format note (fixed 2026-08-08)

Shipped with the retired author identity (Onat Yılmaz / yogt1984@) and a 12pt/lin titlepage layout; both normalized to the house shell in 903264b. Keeps natbib + thebibliography — it is the only paper in the set with inline citations, so author-year stays.

Measurement platform

Rust ingestion engine on the full book + trade stream, **236-dim feature vector per symbol every 100 ms**, 21 categories: book imbalance at multiple depths, raw depth, microstructure dynamics (queue position, imbalance velocity), trade flow / aggressor ratios, VWAP deviation, entropy (permutation, book-shape), Hawkes-type arrival intensity, realized & range volatility, spread and toxicity proxies, funding/OI state, cross-symbol aggregates. **End-to-end feature latency < 80 ms at p99.**

Census window

Three contiguous days, 2026-05-19 to 05-21. 2.17 M ticks/symbol at 100 ms. Spearman rank IC vs future mid returns at six horizons (1 s, 5 s, 30 s, 1 m, 5 m, 15 m; selected features also 100 m) on ~21,708 subsampled points/symbol — **one per ~10 s, deliberately, to control overlap**. Overlapping samples are how this kind of census inflates its own significance; the subsampling is the defence.

Robustness detail

Per-day rolling IC @5 s	BTC +0.433 ± 0.061
	ETH +0.400 ± 0.098
	SOL +0.355 ± 0.071
Bootstrap 95% CI (BTC, [0.442, 0.465], 1,000 resamples	
imbalance_qty_11)	
Time of day	no dead zones; weakest 12–16 UTC, strongest 04–08 UTC
Vol regime (median split)	low +.458/ +.435/ +.391 vs high +.422/ +.394/ +.342

Feature rank ordering preserved across symbols and subwindows — this is what “universal” means operationally.

The one systematic failure

[!] Validation matrix: BTC 7/8, ETH 6/8, SOL 6/8. The failing cell is the *same* one every time — **temporal out-of-sample drift**: IC delta −0.17 across a six-week gap against a 0.10 tolerance. Partially attributable to the efficiency drift documented above, i.e. the failure and one of the findings may be the same phenomenon. Do not report 7/8 without naming which cell fails.

Siblings

#2 is the apparatus paper this sharpens (cross-cite both ways; pick *one* as lead per supervisor) · #6 supplies the breadth mathematics for the 8 axes · #8 is the venue instrument built around the same barrier.